

- support the initiative
- Large-scale deployment goals
- High-resource R&D
- Domestic production of critical raw materials, components and products

After nearly five years of this initiative, with the mission taking off well the target has been revised to 100,000MW solar power generation capacity with the salient features remaining the same.

The way forward

With the threat of climate change, solar is the technology to rely upon in addition to the other forms of renewable energy. Solar electricity generation is one of very few low-carbon energy technologies with the potential to grow to very large scale. Recent years have seen rapid growth in installed solar generation capacity, great improvements in technology, price and performance, and development of creative business models that have spurred investment in residential solar systems.

Nonetheless, further advances are needed to enable a dramatic increase in solar contribution at socially acceptable costs. Achieving this role for solar energy will ultimately require that solar technologies become cost-competitive with fossil generation, which are appropriately penalised for carbon dioxide (CO₂) emissions with substantially reduced subsidies.

Technical nuances

Photovoltaic modules (PVMs). The cost of installed PVM is historically driven by: the cost of the solar module and the balance-of-system (BOS) costs, which include costs of inverters, racking and installation hardware, design and installation labour, and marketing, as well as various regulatory and financing costs. After years of development supported by the government's research and development investments, the leading PVM technology, termed wafer-based crystalline silicon (c-Si), is

technologically mature with large-scale c-Si module manufacturing capacity in place. c-Si systems are likely to dominate the solar energy market for the next few decades and perhaps beyond. Moreover, if the industry can substantially reduce its reliance on silver for electrical contacts, material inputs for c-Si PV generation are available in sufficient quantity to support expansion to terawatt scale.

However, current c-Si technologies have inherent technical limitations—most importantly, their high processing complexity and low intrinsic light absorption (which requires a thick silicon wafer). The resulting rigidity and weight of glass-enclosed c-Si modules contribute to BOS cost. Firms that manufacture c-Si modules and their component cells and input materials have the means and the incentive to pursue remaining opportunities to make this technology more competitive through improvements in efficiency and reductions in manufacturing cost and materials use. Thus, there is not a good case for government support of R&D on current c-Si technology.

The limitations of c-Si have led to research into thin-film PVM alternatives. Commercial thin-film PVM technologies, primarily cadmium telluride (CdTe) and copper-indium gallium-selenide (CIGS) solar cells, constitute roughly 10 per cent of the PVM market today and are already cost-competitive with silicon. Unfortunately, some commercial thin-film technologies are based on scarce elements, which makes it unlikely that they will be able to achieve terawatt-scale deployment at reasonable cost. The abundance of tellurium in the earth's crust, for example, is estimated to be only one-quarter that of gold.

Emerging thin-film technologies, which are in the research stage, use neo-age material systems and device structures and have the potential to provide superior performance with lower manufacturing complexity and

cost. Several of these technologies use materials abundant on the earth—even silicon in some cases. Other properties of some new thin-film technologies, such as low weight and compatibility with installation in flexible formats, promise reductions in BOS costs along with lower module costs.

Though these emerging technologies are not nearly competitive with c-Si today, they have the potential to significantly reduce the cost of PVM-generated electricity in the future. And while the private sector is likely to view R&D investments in these technologies as risky, the payoff could be enormous. Therefore, to increase the contribution of solar energy for long-term climate change mitigation, I strongly recommend that a large fraction of government resources for solar research and development focuses on environmentally benign, emerging thin-film technologies that are based on Earth-abundant materials. Technically, that would be the ideal way forward, catapulting India into a new realm of growth by making it energy powered!

Need for solar powered cities

With the changing urban landscapes and rapid economic development, we are seeing a spurt in energy demand of urban centres, leading to enhanced greenhouse gas (GHG) emissions. Mega cities across the world have set up stiff goals to be renewable-energised and reduce GHG, and countries like USA, China and Australia have already started setting up Solar Cities. So is the time ripe for India to follow the same?

Indian cities and towns are experiencing rapid growth in their peak electricity demand. The local governments and the electricity utilities are finding it difficult to cope with this rapid rise in demand. As a result, most of the cities/towns are facing electricity shortages.

